CFD Analysis of Different Rear Wing Endplate Geometries for Racing Vehicles and Their Effect on Drag and Downforce

Rhiana Boutot[1], Alexis Reiff[2], Jack Rollinson[3], Leopold W. Zitzmann[4]
Wentworth Institute of Technology, Boston, MA, 02115, United States of America

Faculty advisor Professor El-Sadi

High-speed racing vehicles rely on aerodynamic devices to reduce lift, increase tire loading, and maintain stability during straight-line travel and cornering. Rear wings generate negative lift, or downforce, without substantially increasing vehicle mass; however, the pressure difference between the upper and lower surfaces of the wing produces wingtip vortices that contribute to induced drag. Endplates reduce pressure-driven airflow around the wing tips and can incorporate additional geometric features to further control vortex formation and wake behavior. This paper investigates the aerodynamic performance of four rear-wing endplate configurations: a reference endplate, slats, cutouts, and strakes. Each configuration is modeled and evaluated using computational fluid dynamics to examine lift, drag, pressure distribution, velocity, and airflow trajectories. Mesh refinement was applied to improve the reliability and consistency of the numerical comparison. Three-dimensional printed prototypes were subsequently tested in a wind tunnel at angles of attack of 0°, 10°, and 15°, allowing the experimental results to be compared with the computational analysis. The results demonstrate that endplate geometry and angle of attack significantly affect the balance between downforce generation and aerodynamic drag. Slatted endplates produced the greatest downforce-to-drag ratio at 10° and 15°, while strakes generated the greatest absolute downforce at 15°. These findings demonstrate the value of combining simulation-driven design with experimental validation when developing rear-wing endplates for improved aerodynamic efficiency and vehicle stability.

Nomenclature

A_f	=	Frontal Area
A_p	=	Projected area
C_D	=	Drag coefficient
C_L	=	Lift coefficient
D	=	Drag force
L	=	Lift force
V	=	Air velocity
ρ	=	Air density

[1] Undergraduate Mechanical Engineering, School of Engineering
[2] Undergraduate Mechanical Engineering, School of Engineering
[3] Undergraduate Mechanical Engineering, School of Engineering
[4] Undergraduate Mechanical Engineering, School of Engineering, Student Member (1926031).

I. Introduction

High-speed car racing competitions combine high-speed thrills and great scientific innovation. These cars are feats of engineering since racers gauge aerodynamics to increase performance, efficiency, and safety. Although there is limited information available to the public about specific racecar considerations due to the nature of the competition, there are general aerodynamic concepts that can be analyzed to better understand how racers optimize their efficacy. Lift forces, for example, are introduced to vehicles as the open-wheel cars move at faster velocities [1]. Racecars experience considerable amounts of these lift forces due to the nature of the sport; they can become dangerous as it decreases the grip tires have on the road, destabilizing the cars during high speed straight-aways and corner turns. On the other hand, high-speed cars must be lightweight to move fast and use fuel efficiently. How is it possible to reduce these lift forces without increasing the weight of vehicles? High-speed racecars utilize wings, and more specifically rear wings, to counteract lift forces [2].

Rear wings were inspired by the aeronautical field to counteract lift and drag forces without adding too much weight to the racecars. They are mostly made of composite materials, such as carbon fiber composites, as they are lightweight, yet strong materials [3]. A myriad of different airfoil shapes and angles of attack are utilized by rear wings to best offset lift and drag forces by generating thousands of newtons or pounds of downforce. Rear wings were very successful when introduced to racecars, leading to the addition of wings to everyday automobiles to help increase fuel efficiency, safety, and stability. Although, the main objective for this experiment will focus on rear wings' effect on racing vehicles.

Rear wings and aerodynamic add-on devices have been used in automotive applications since the early development of high-speed racing vehicles. Some of the earliest versions, including front underbody dams and rear deck lips, appeared in the early 1960s on race cars, where their primary purpose was to assist in reducing aerodynamic lift and help maintain vehicle stability at increasing speeds. As vehicle speeds increased, especially in racing, stability and drag became more important design concerns, which led to the greater use of rear wings on racing vehicles in the late 1960s. By the 1970s, rear wings became commonly used in motorsport applications, and eventually started to appear on regular passenger cars as well. Their main goal within those vehicles was to improve fuel efficiency by reducing drag at higher driving speeds. However, not all rear wing endplate implementations improved aerodynamic performance, as some were introduced primarily for stylistic choices and could negatively impact airflow performance when poorly designed. [4]

As rear wing designs developed over time, engineers started to improve their geometry to better manage any airflow separation at the back of the vehicle. This design choice is necessary since this region of the car strongly influences the overall drag. In modern applications, additional features have been used to help control airflow near the edges of the rear wings and reduce vortices that contribute to increased drag, such as endplates. The improvements to the wing design have been studied and optimized using CFD and experimental testing. This allows engineers to have a greater understanding of how rear wing geometry can have a direct correlation on the effect of drag performance on both racing and passenger vehicles. [5] [6]

A fundamental challenge in rear wing aerodynamics is the formation of wingtip vortices caused by the pressure difference between the upper and lower surfaces of the wing. In inverted racing airfoils designed to generate downforce, low pressure develops beneath the wing while high pressure forms along the upper surface. Near the wing tips, airflow naturally moves from the high-pressure region toward the low-pressure region, causing the flow to roll into strong vortex generations known as wingtip vortices. These vortices induce rotational energy into the wake flow and contribute significantly to induced drag. Unlike many conventional aeronautical applications where viscous drag accounts for a large portion of total aerodynamic loss, motorsport aerodynamics places a strong emphasis on downforce generation, making induced drag a major performance consideration. Therefore, reducing vortex strength while maintaining or increasing downforce is critical in the design of a competitive rear wing [7].

To reduce the aerodynamic penalties associated with wingtip vortices, endplates are commonly employed at the outer edges of rear wings. Endplates act as physical barriers that reduce airflow spillover between the high-pressure region above the wing and the lower-pressure region beneath it, weakening vortex formation and improving aerodynamic efficiency. By preserving the pressure differential across the wing surfaces, endplates allow the wing to generate greater downforce while reducing induced drag. Modern endplate designs often incorporate geometric features such as louvers, slats, cutouts, and strakes to further manipulate airflow structures and wake behavior. For example, louvers allow high-pressure air to bleed outward from the inner region of the endplate to the surrounding flow, weakening vortex intensity and reducing drag but sacrificing downforce. Similarly, cutouts have been proposed to generate secondary vortices that interact with the primary wingtip vortex, potentially reducing wake strength and improving overall aerodynamic performance [8].

To investigate these aerodynamic effects, this project focuses on the design and analysis of multiple rear wings and endplate configurations using both computational and experimental methods. Several three-dimensional models will be developed in SOLIDWORKS with varying geometric features intended to manipulate airflow structures surrounding the rear wing assembly. Each design will undergo computational fluid dynamics (CFD) analysis to compare the aerodynamic characteristics of drag force and lift. Additionally, each design will be fabricated using additive manufacturing techniques. Experimental testing will then be conducted in a wind tunnel environment to further evaluate the aerodynamic behavior of the selected design and compare the experimental results with the computational analysis. [9]

Computational fluid dynamics has become a fundamental tool in modern aerodynamic research due to its ability to model and predict fluid flow behavior around complex geometries. CFD utilizes numerical methods and governing fluid mechanics equations to simulate airflow interactions Within motorsport engineering, CFD is extensively utilized because it allows engineers to rapidly analyze multiple design iterations without requiring immediate full-scale physical prototypes or repeated wind tunnel testing. This capability significantly reduces development time and cost while providing detailed visualization of airflow behavior surrounding aerodynamic components. CFD is particularly valuable for analyzing rear wing and endplate configurations because small geometric modifications can substantially affect downforce production and wingtip vortex generation. For this experiment, CFD simulations will be performed on each rear wing model to evaluate aerodynamic efficiency and identify the configuration that produces the optimal balance between drag reduction and downforce generation prior to experimental validation in the wind tunnel. [10] [11]

II. Solid works model

The first iteration of the endplate design includes the addition of three louvers near the upper rear corner of the endplate, located approximately 70-80% of the wing chord length from the leading edge, shown in Figure 1. This location was selected because wingtip vortices become stronger toward the trailing edge due to the pressure difference across the wing. The louvers are angled to align with the local airflow while providing controlled pressure relief across the endplate. They accomplish this by allowing a controlled amount of high-pressure air to pass through the endplate, reducing the pressure differential across it and weakening the wingtip vortex. This design aims to reduce induced drag and improve the aerodynamic efficiency of the rear wing.

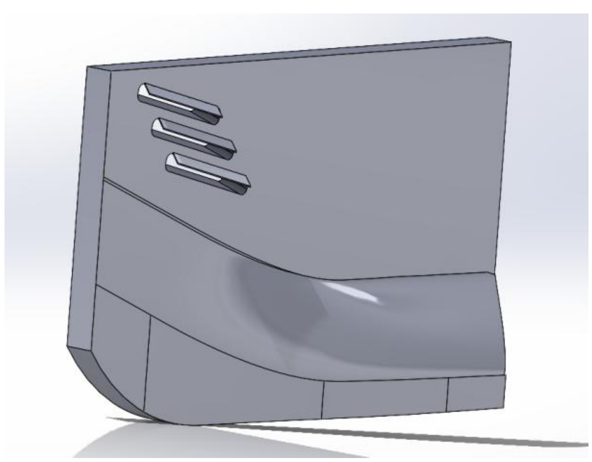

Fig 1. Endplate with Louvers

The second iteration of the endplates involves the addition of slats near the rear of the endplates and on the low-pressure side of the rear wing, shown in Figure 2. These slats are designed to influence the airflow exiting the diffuser and underbody region rather than only managing the flow around the wing itself. By guiding the diffuser wake upward toward the rear wing, the slats can help energize the flow, reduce separation, and improve the interaction between the diffuser and wing. This can increase the overall pressure difference across the rear wing and improve downforce production. However, because the slats introduce additional surfaces and flow disturbances, the increase in downforce may come with a corresponding increase in drag. Therefore, this geometry represents a tradeoff between improved aerodynamic load and reduces aerodynamic efficiency.

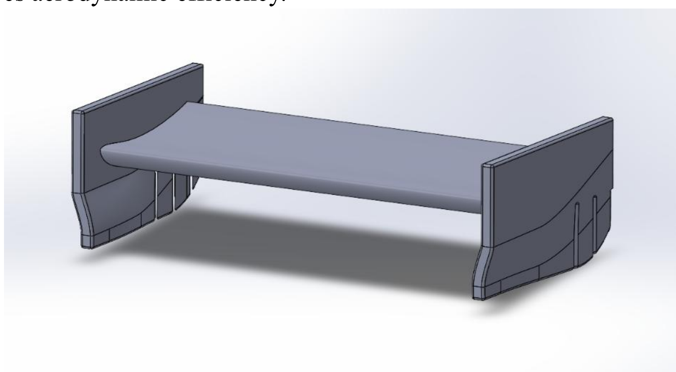

Fig 2. Endplate with Slats

The third iteration of the endplates incorporates the addition of stepped cutouts out of the top rear of the endplates, shown in Figure 3. Endplate cutouts generally decrease drag force. A vortex is created on the trailing edge of the endplates which rotates in the opposite direction of the vortex created by the wingtip of the airfoil, thus theoretically reducing drag. The third iteration includes a stepped design, which is based off a myriad of Formula 1 open-wheel racecar rear wings. However, the stepped design also includes fillets to reduce stress on sharp corners. Multiple professional racecar rear wings also include a curved design instead of steps. The decision to combine both types of cutouts was to optimize the geometry, since the vortices created by the trailing edge largely depend on the cutout geometry.

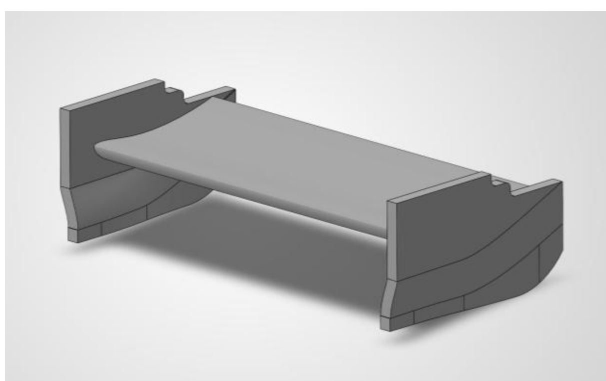

Fig 3. Endplate with Cutouts

The fourth iteration of the endplates adds the additional vertical strakes positioned along the rear section of the endplate geometry, shown in Figure 4. The extension increases the surface area available to influence the airflow near the trialing edge, where the wingtip vortices become more dominant due to the pressure difference across the wing. The strakes are positioned to organize and redirect the airflow, generating controlled streamwise vortices that help stabilize the wake behind the rear wing. By limiting airflow migration between high- and low-pressure regions, the design reduces vortex strength and improves pressure retention across the wing surfaces. This configuration aims to increase downforce generation while minimizing induced drag.

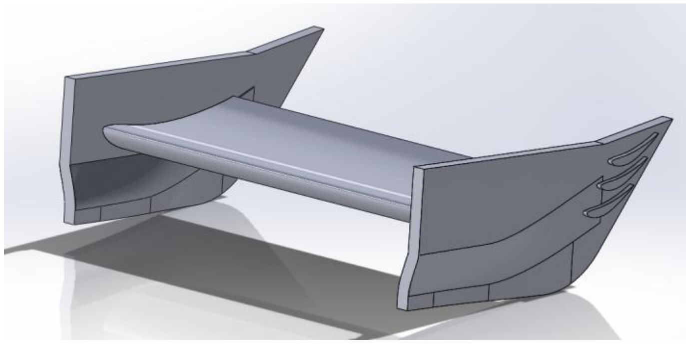

Fig 4. Endplate with Strakes

III. CFD Analysis

Computational fluid dynamics (CFD) simulations were performed in SolidWorks Flow Simulation to compare the aerodynamic performance of the four rear wing endplate geometries. The simulations modeled three-dimensional external airflow around each rear wing using air as the working fluid. Each geometry was analyzed under the same operating conditions so that any differences in drag, lift, downforce, pressure, and velocity were due to the changes in the endplate geometry. The CFD analysis was used to compare the airflow around each design and evaluate the effect of the different endplate geometries on the aerodynamic performance of the rear wing.

The boundary conditions for the CFD simulations were based on the dimensions of the wind tunnel used during testing. A reference coordinate system with an origin at (0,0,-5) was created to position the rear wing so the airflow passed over it correctly during the simulation. The computational domain was also based on the wind tunnel dimensions, and an inlet velocity of 98.4252 ft/s was used to match the wind tunnel test speed. Since the rear wing models were 3D printed, a surface roughness was included in the simulations to better represent the printed parts. The boundary condition used in the simulations are shown in Table 1, and the computational domain dimensions are shown in Table 2.

Reference axis	Inlet velocity	Surface roughness
x-axis	98.4252 ft/s	2×10^{-6} in

Table 1. Physical Boundary Conditions

Right wall (x-direction)	Left wall (x-direction)	Top wall (y- direction)	Bottom wall (y-direction)	Front wall (z-direction)	Back wall (z- direction)
17.014 in	17.014 in	6.000 in	6.000 in	6.000 in	6.000 in

Table 2. Computational Domain Conditions

A. Meshing

Meshing						
Model	Angle °	Total Cells	Fluid Cells	Contacting Fluid Cells	Shape	Iterations
Reference	0	1634127	1634127	42142	Rectangle	236
						269
	10	1656803	1656803	58466	Rectangle	237
						275
	15	1663064	1663064	54519	Rectangle	237
						309
Slats	0	2084160	2084160	242183	Rectangle	256
						290
	10	2147853	2147853	290893	Rectangle	259
						328
	15	2133500	2133500	278153	Rectangle	258
						258
Strakes	0	1585575	1585575	22234	Rectangle	234
						277
	10	1592546	1592546	23482	Rectangle	234
						234
	15	1588270	1588270	23154	Rectangle	234
						253
Cutouts	0	1602667	1602667	19180	Rectangle	235
						303
	10	1605215	1605215	19516	Rectangle	255
						271
	15	1607981	1607981	23537	Rectangle	235
						261

Table 3. Meshing Conditions

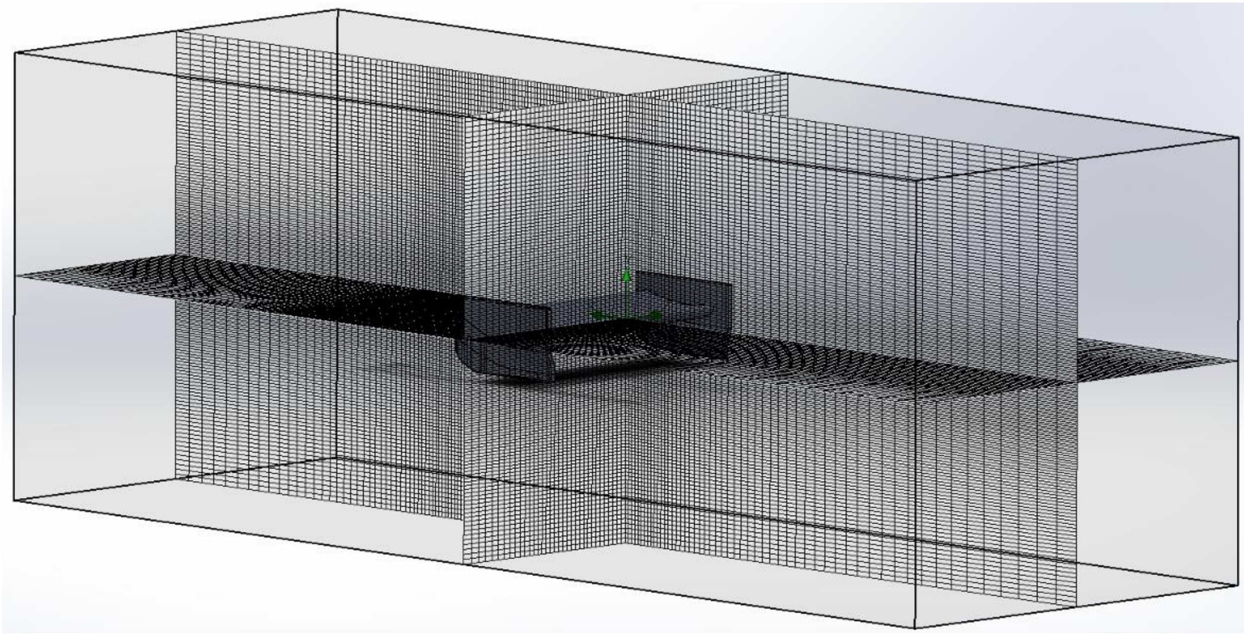
Figure 5. Meshing Shown on Model

B. Analysis

Because of the reference rear wing endplate being evaluated at three varying angles of attack, the goal plots of drag and lift were monitored across the duration of each simulation, and while some minor fluctuations were present within the initial iterations, all three models settled shortly after and demonstrated a satisfactory convergence. This led to a consistent set of results in which the pressure contours across the 0°, 10°, and 15° angles of attack remained relatively similar to one another, with only small changes occurring within the pressure distribution as the angle was increased. The velocity contours and the flow trajectory both portray the airflow being redirected further around the rear wing at 10° and 15°, with small vortices appearing to form near the rear wing endplate at 0°, becoming more visible and apparent at 10°, and showing the greatest increase in size at 15° where the trajectories follow a more curved path around the rear wing endplates. With increasing angles of attack, the reference endplate redirects a greater portion of the flow around itself while the vortices grow in size, evident of the direct effect the angle of attack has on the airflow. The solution was determined to remain converged at all three angles, indicating that the changes observed across the contours and trajectories are consistent and reputable results.

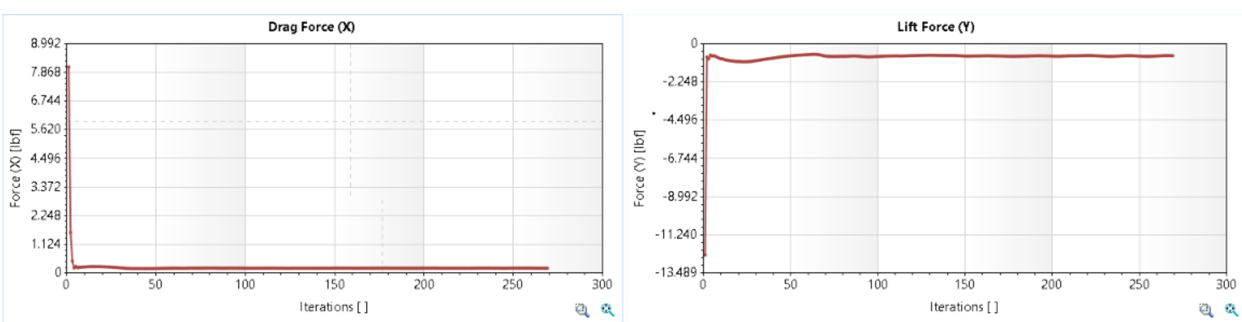

Figure 6. Reference, 0 Degrees Angle of Attack, Drag and Lift Forces vs. Iterations

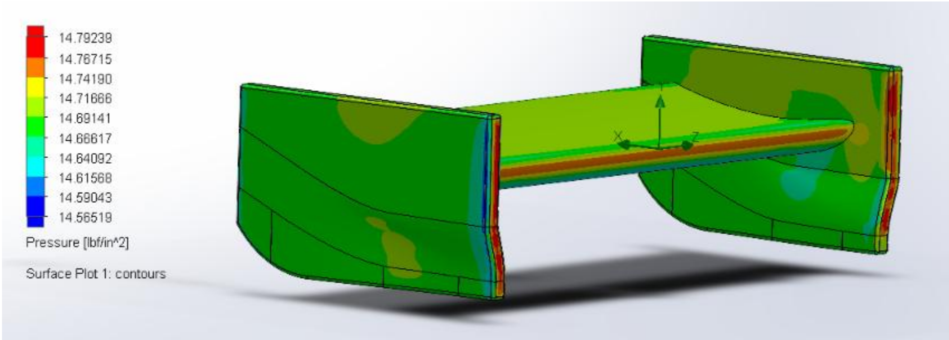

Figure 7. Reference, 0 Degrees Angle of Attack, Pressure Surface Plot

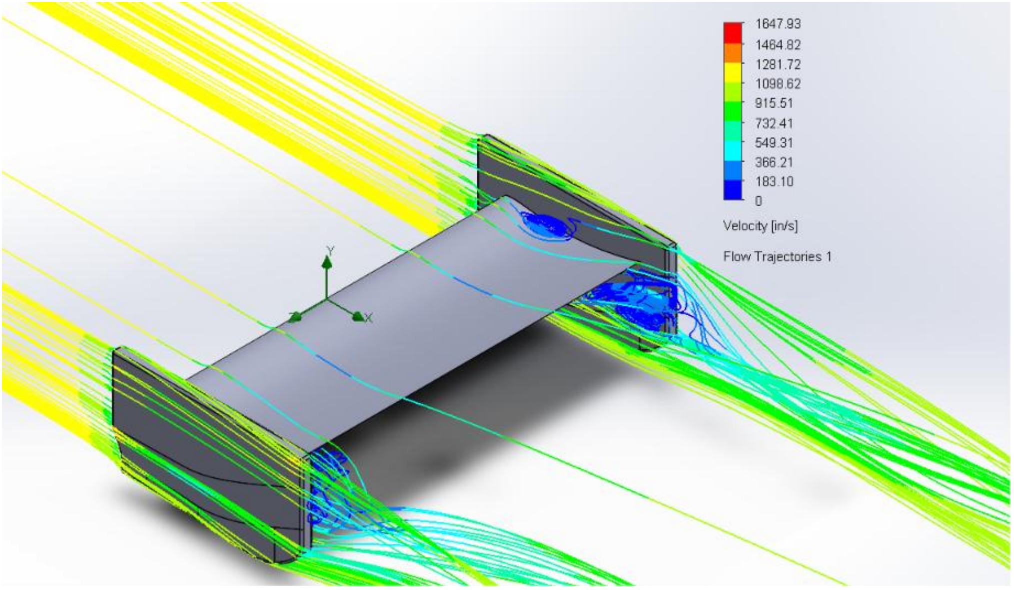

Figure 8. Reference, 0 Degrees Angle of Attack, Velocity Flow Trajectories

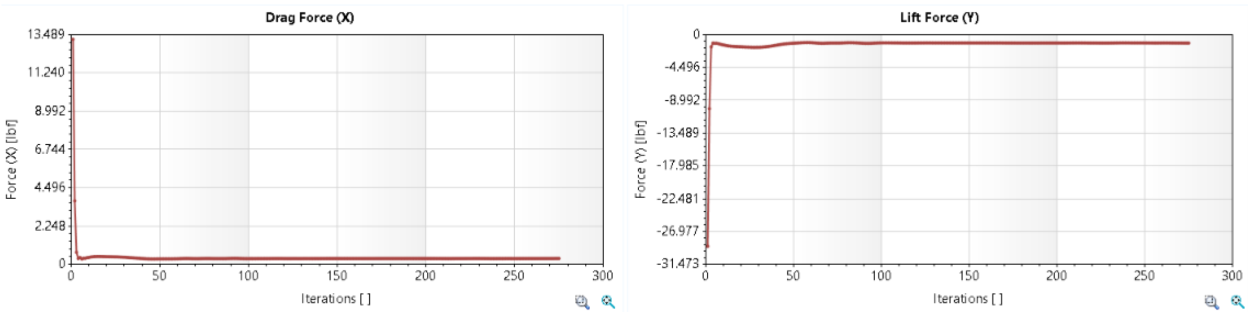

Figure 9. Reference, 10 Degrees Angle of Attack, Drag and Lift Forces vs. Iterations

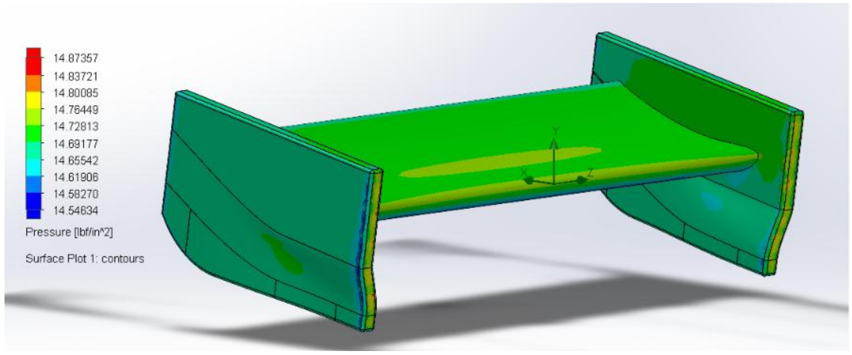

Figure 10. Reference, 10 Degrees Angle of Attack, Pressure Surface Plot

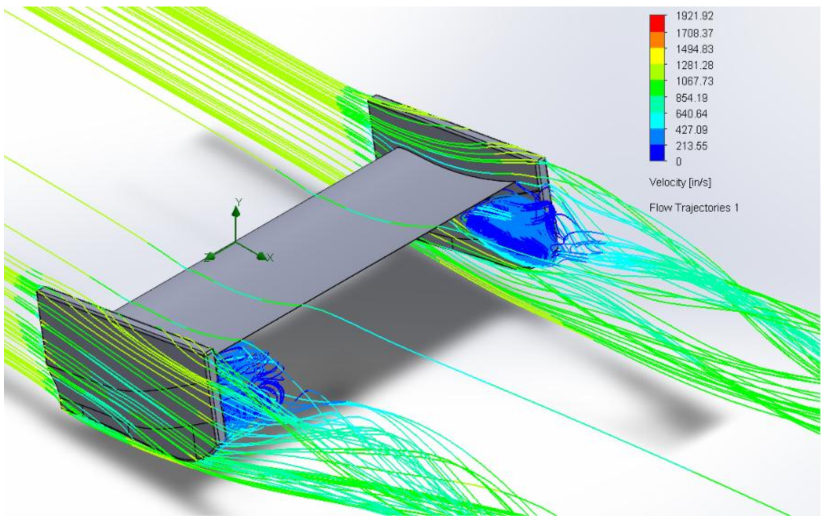

Figure 11. Reference, 10 Degrees Angle of Attack, Velocity Flow Trajectories

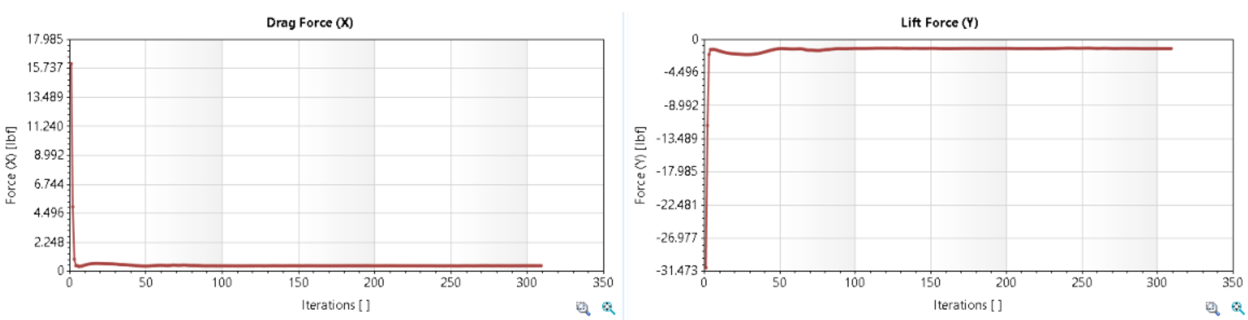

Figure 12. Reference, 15 Degrees Angle of Attack, Drag and Lift Forces vs. Iterations

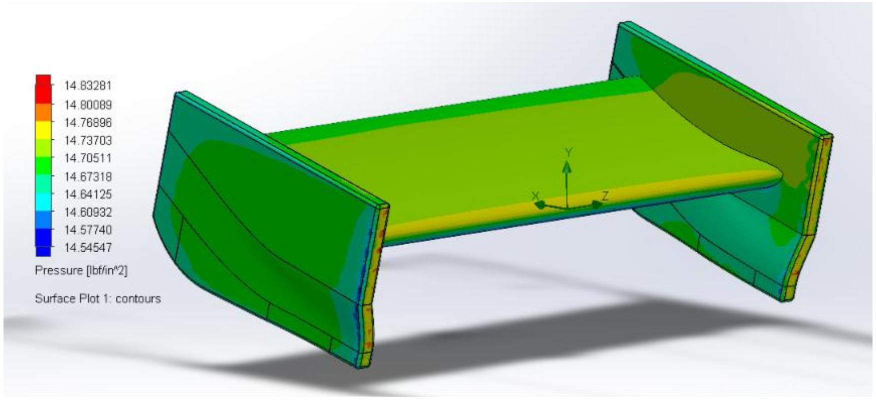

Figure 13. Reference, 15 Degrees Angle of Attack, Pressure Surface Plot

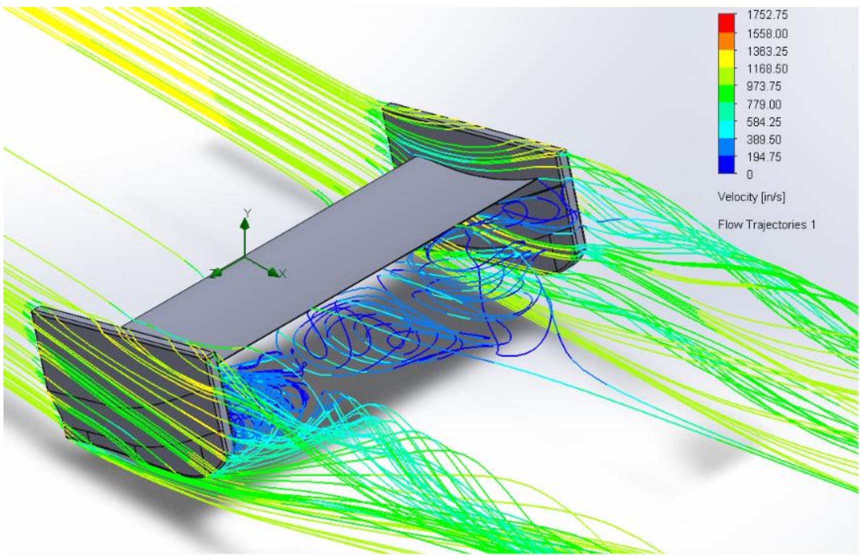

Figure 14. Reference, 15 Degrees Angle of Attack, Velocity Flow Trajectories

The rear wing endplate model with incorporated strakes edge at the three varying angles of attack all demonstrated convergence throughout the duration of the simulation. The goal plots of drag and lift shows minor fluctuations during the initial iterations before stabilizing, which is to be expected under these conditions. The pressure contours at 0°, 10°, and 15° angle of attack remained relatively similar, although the pressure across the rear wing became more apparent as the angle of attack was increased, particularly near the leading edge of the airfoil and the sides of the winglets. The velocity contours portray that the strakes increasingly redirect and organize the airflow around the rear wing as the angle of attack increases. At 0°, only small vortices form near the strakes and the flow remains relatively uniform. At 10°, the vortices become more visible, and the flow trajectories begin to follow a more curved path around the endplate. At 15°, the vortices exhibit the greatest size and strength, with the airflow remaining attached to the strakes further downstream producing a more organized flow. While this configuration may provide the most downforce it also produces the most amount of drag. The results point to an increase in angle of attack strengthens the aerodynamic influence of the strakes by increasing the pressure and creating a larger vortex.

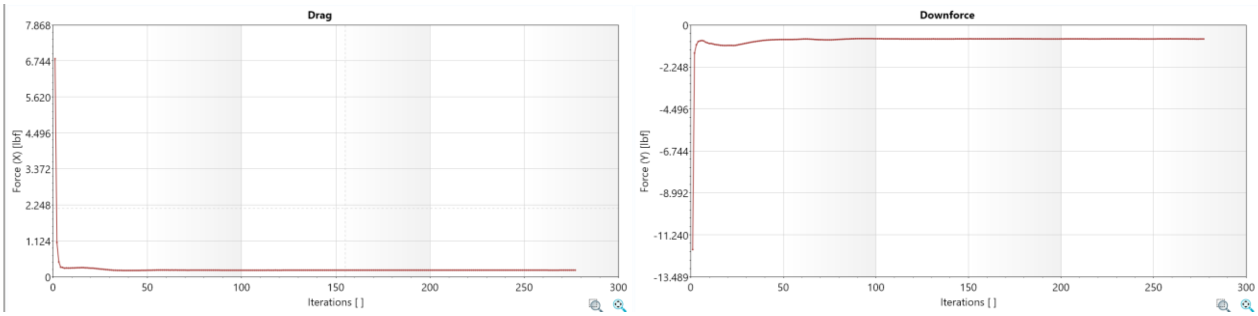

Figure 15. Strakes, 0 Degrees Angle of Attack, Drag and Lift Forces vs. Iterations

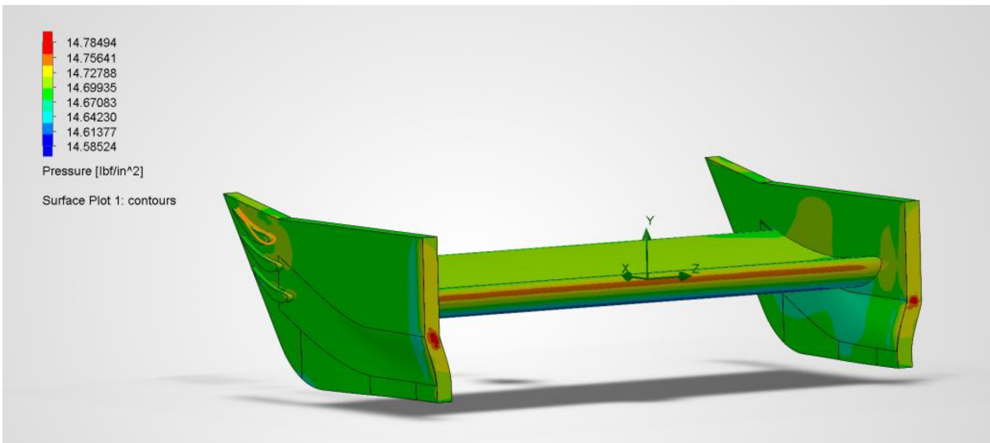

Figure 16. Strakes, 0 Degrees Angle of Attack, Pressure Surface Plot

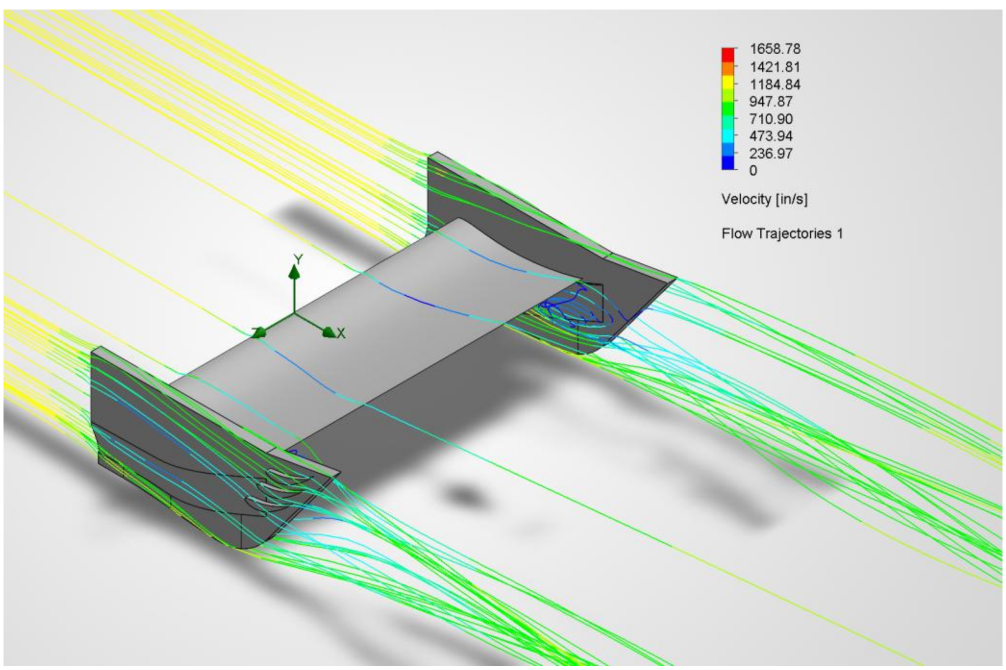

Figure 17. Strakes, 0 Degrees Angle of Attack, Velocity Flow Trajectories

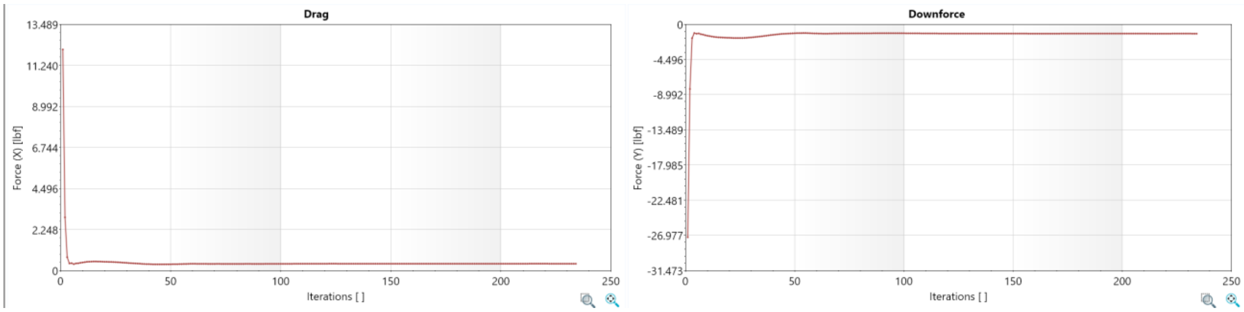

Figure 18. Strakes, 10 Degrees Angle of Attack, Drag and Lift Forces vs. Iterations

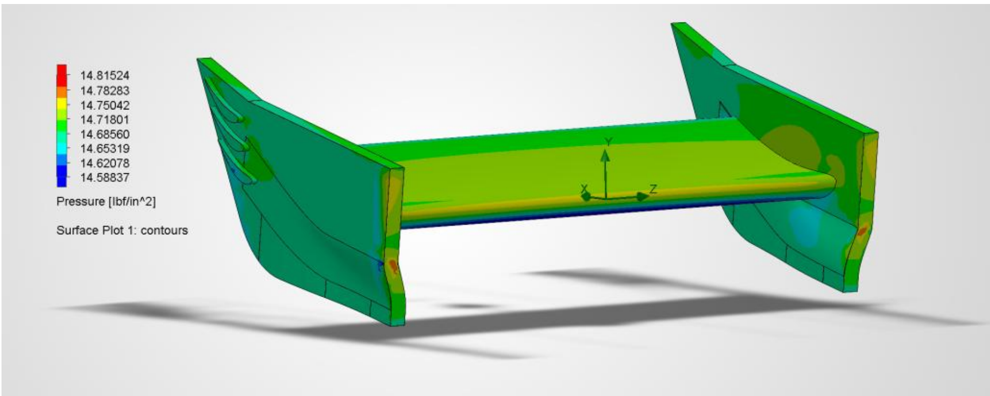

Figure 19. Strakes, 10 Degrees Angle of Attack, Pressure Surface Plot

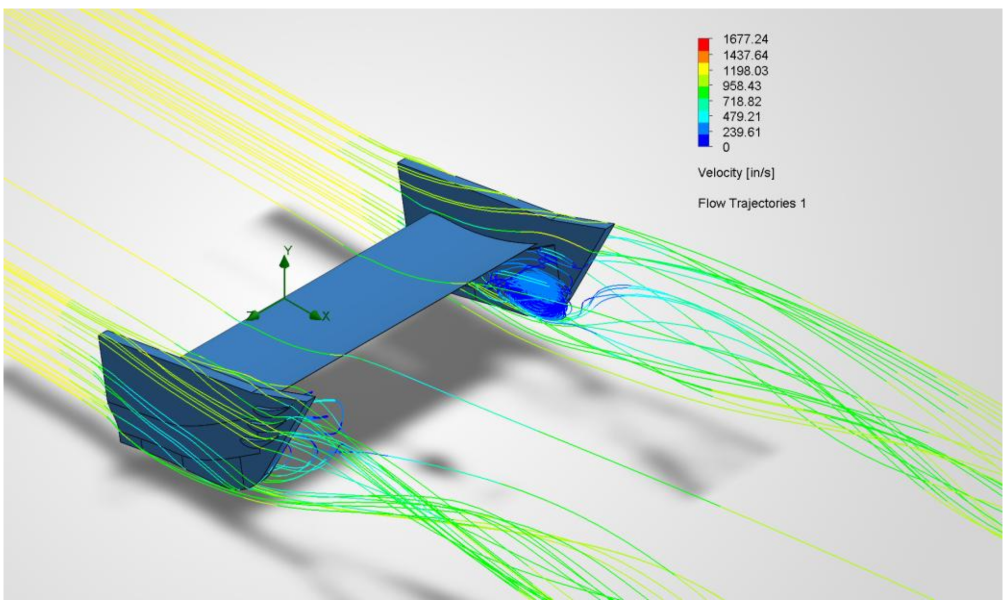

Figure 20. Strakes, 10 Degrees Angle of Attack, Velocity Flow Trajectories

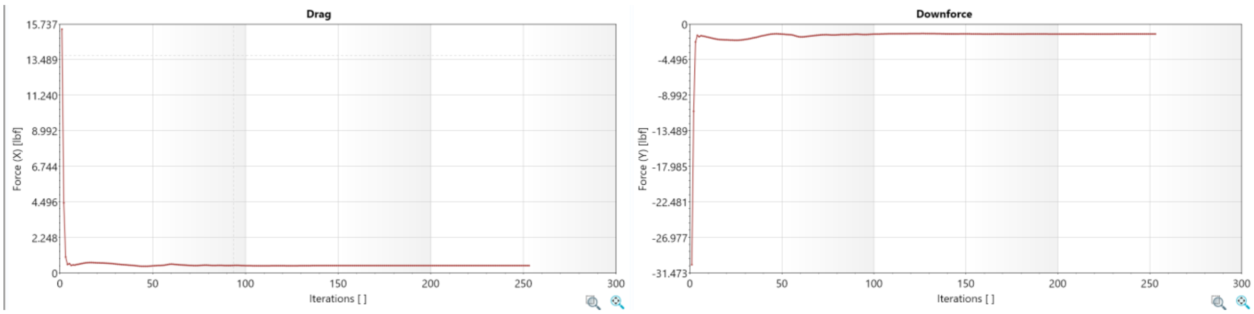

Figure 21. Strakes, 15 Degrees Angle of Attack, Drag and Lift Forces vs. Iterations

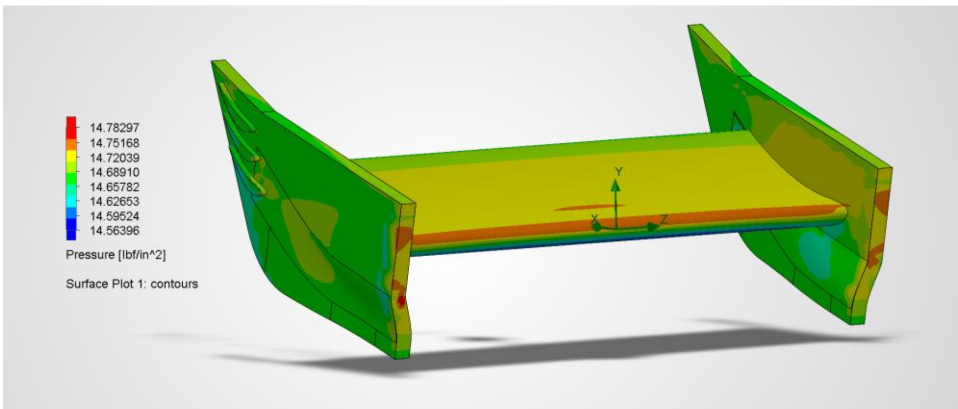

Figure 22. Strakes, 15 Degrees Angle of Attack, Pressure Surface Plot

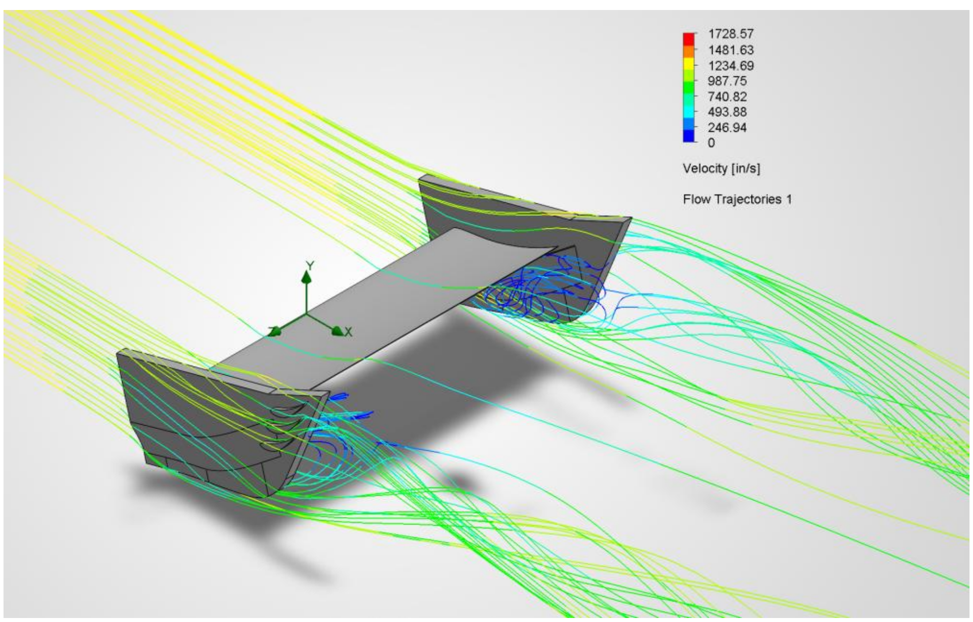

Figure 23. Strakes, 15 Degrees Angle of Attack, Velocity Flow Trajectories

Due to the geometry of the slats requiring fluid to flow between the walls of the slats, more fluid cells were required for the mesh to generate an accurate analysis given the same mesh refinement as the other endplates. This led to a slightly more detailed look of the flow trajectory with air being drawn from the low-pressure side of the endplate toward the high-pressure side. With increasing angles of attack, the slats change the vortex into a larger, more structured flow pattern while preserving downforce and reducing overall drag, evident of the increase in aerodynamic efficiency. The ideal angle of attack for the slats was determined to be 10°, indicating that the slats work to lessen the effect of the vortices coming off the airfoil.

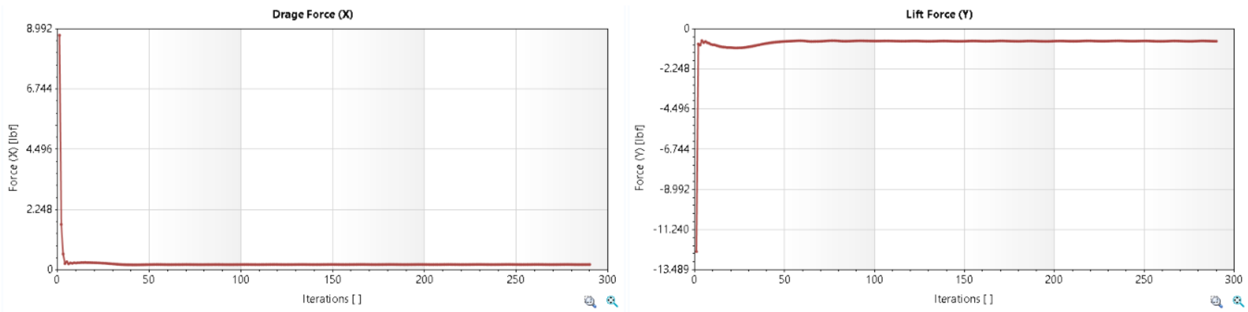

Figure 24. Slats, 0 Degrees Angle of Attack, Drag and Lift Forces vs. Iterations

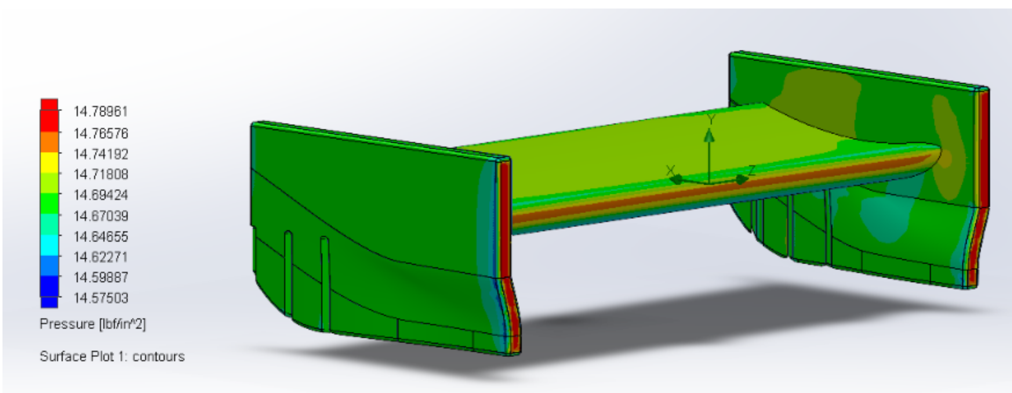

Figure 25. Slats, 0 Degrees Angle of Attack, Pressure Surface Plot

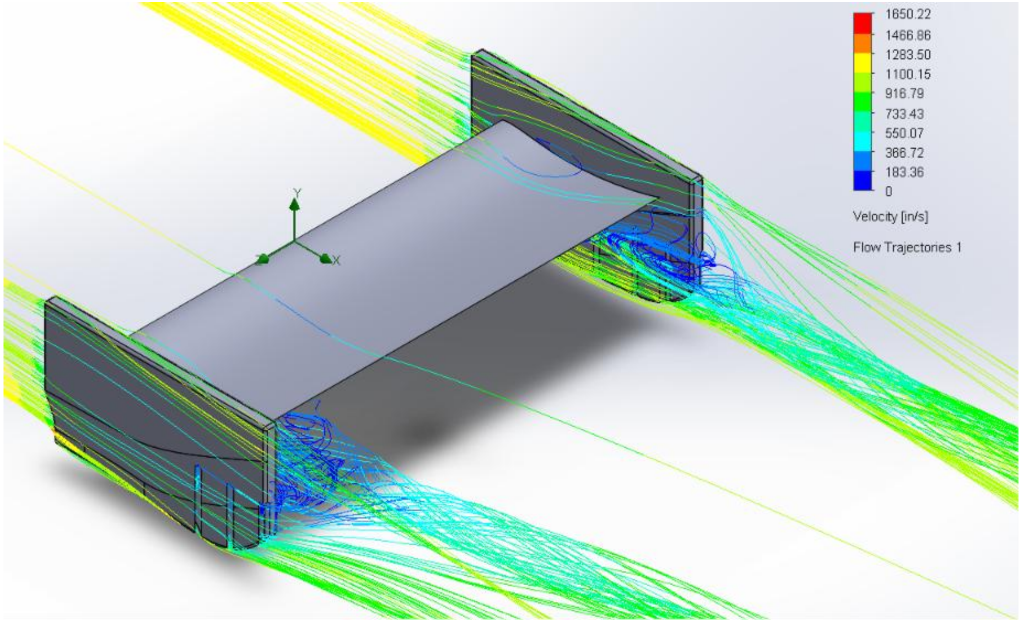

Figure 26. Slats, 0 Degrees Angle of Attack, Velocity Flow Trajectories

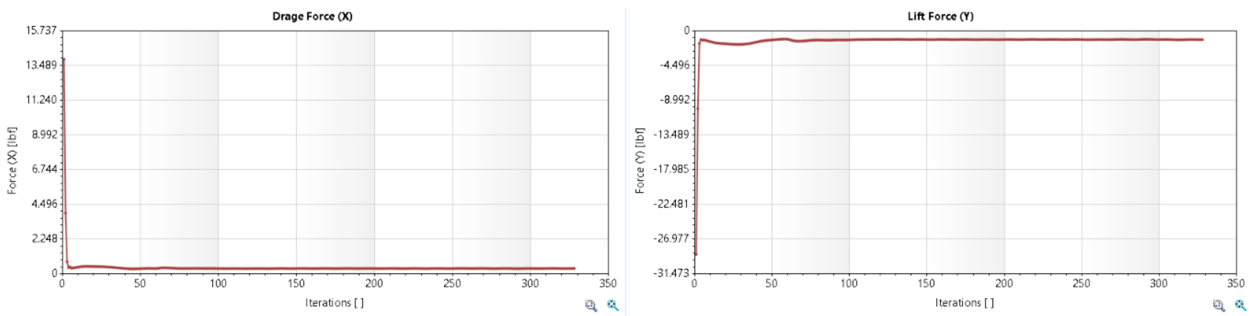

Figure 27. Slats, 10 Degrees Angle of Attack, Drag and Lift Forces vs. Iterations

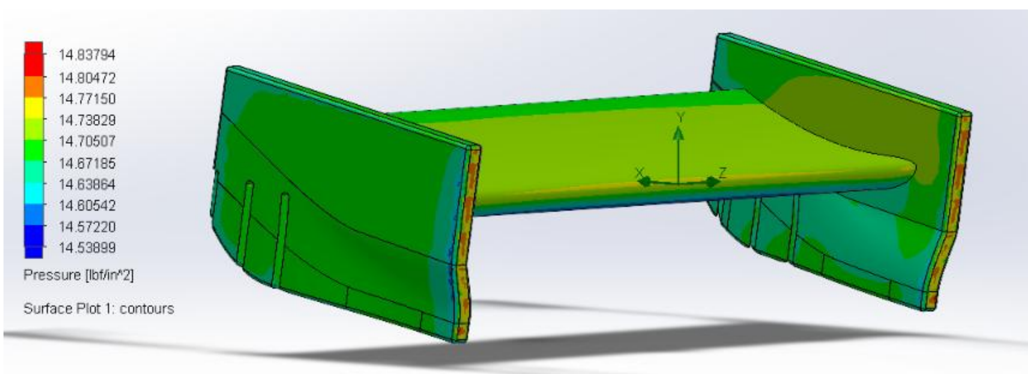

Figure 28. Slats, 10 Degrees Angle of Attack, Pressure Surface Plot

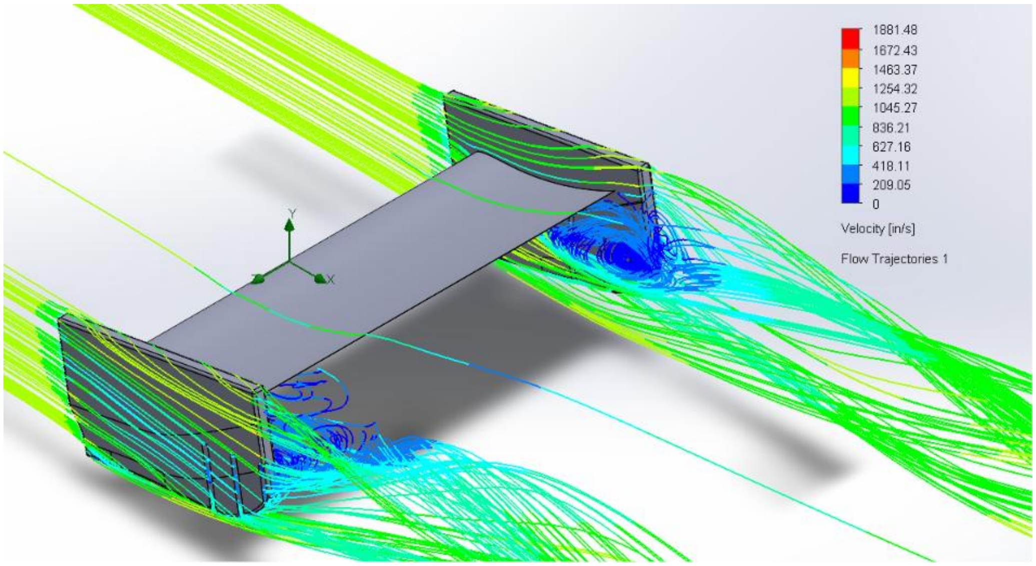

Figure 29. Slats, 10 Degrees Angle of Attack, Velocity Flow Trajectories

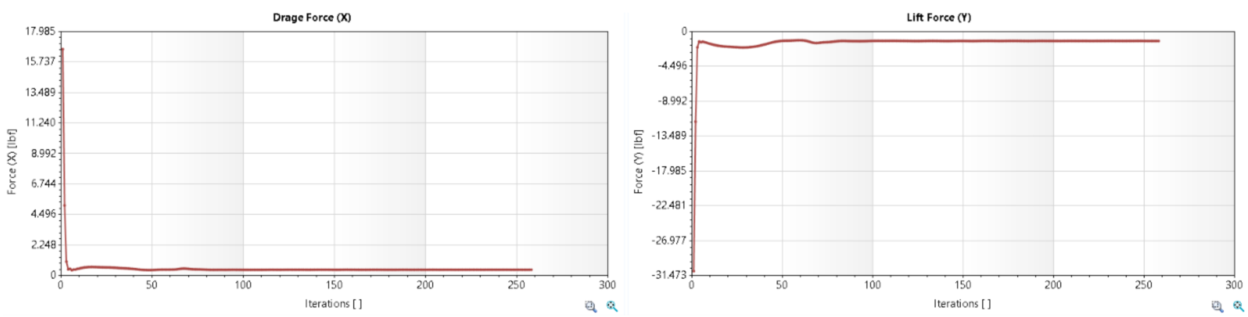

Figure 30. Slats, 15 Degrees Angle of Attack, Drag and Lift Forces vs. Iterations

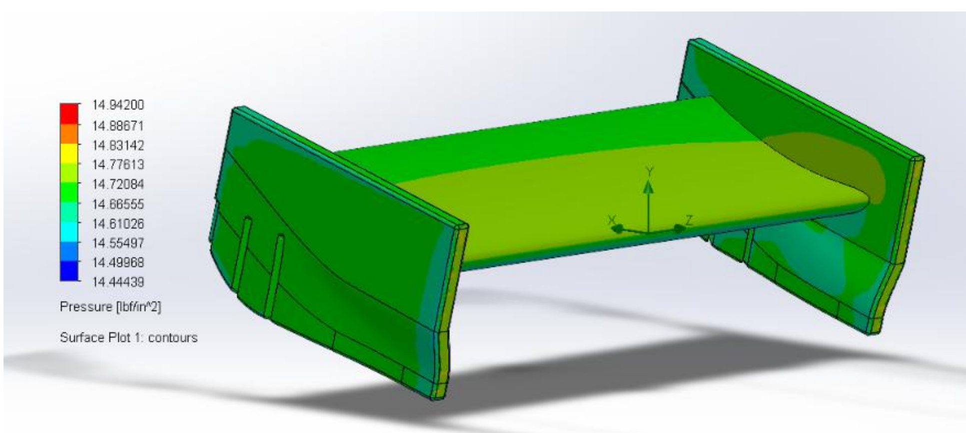

Figure 31. Slats, 15 Degrees Angle of Attack, Pressure Surface Plot

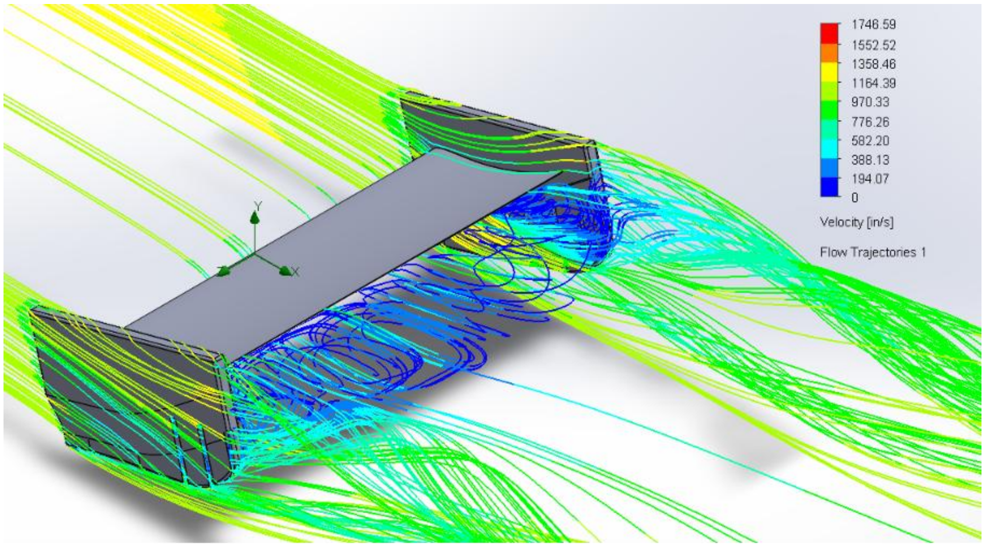

Figure 32. Slats, 15 Degrees Angle of Attack, Velocity Flow Trajectories

The model incorporating cutouts on the trailing edges of the endplates was examined at three different angles of attack: 0°, 10°, and 15°. These three simulations evaluated the drag and downforces that each model configuration achieved with air moving at 1181.1 in/s towards the leading edges of the endplates. As seen below in Figures 33, 36, and 39, each simulation successfully converged & drag and downforces were determined for each simulation. Each simulation underwent roughly 250 iterations to reach convergence. The three simulations also show similar pressure contours, where there is an area of high pressure near the top-front of the airfoil, and an area of low pressure on the bottom of the airfoil. This is to be expected since high pressure is necessary on top of the airfoil to generate downforce on the design. The only difference between each configurations' pressure contours is that the pressure differential between the top and bottom of the airfoil increases with the increased angle of attack. Further, velocity flow trajectories are shown for each configuration and again, they are all relatively similar and increase in magnitude as the angle of attack increases. The vortices coming off the trailing edge of the endplates are shown in the flow trajectories. Although the hypothesis for cutouts claims that these trailing edge vortices should counteract the airfoil vortices, that is not necessarily transparent in the three configurations below. This could be due to the small scale of the model and simulation, as well as the fact that the simulation does not take the rest of the car into account, unlike the hypothesis.

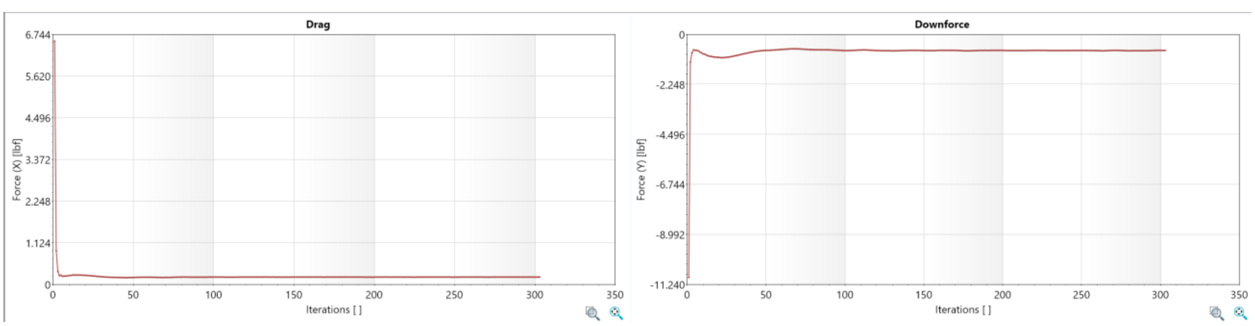

Figure 33. Cutouts, 0 Degrees Angle of Attack, Drag and Lift Forces vs. Iterations

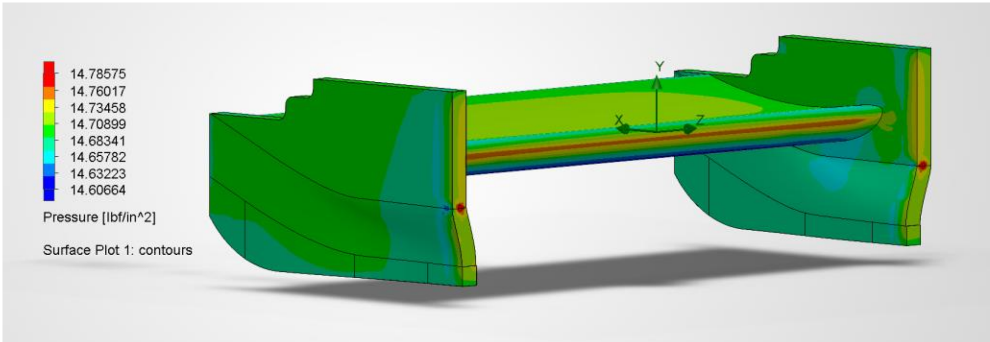

Figure 34. Cutouts, 0 Degrees Angle of Attack, Pressure Surface Plot

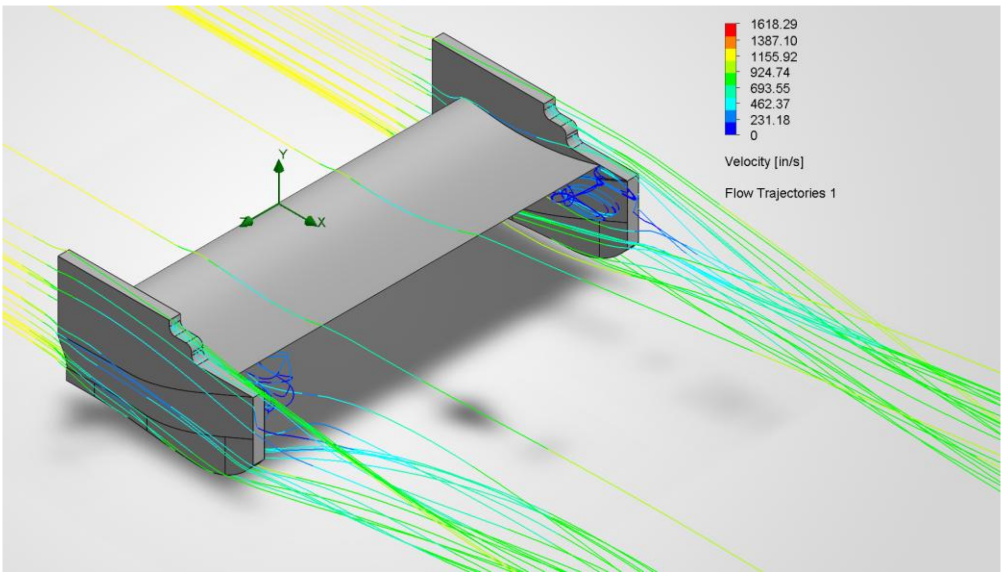

Figure 35. Cutouts, 0 Degrees Angle of Attack, Velocity Flow Trajectories

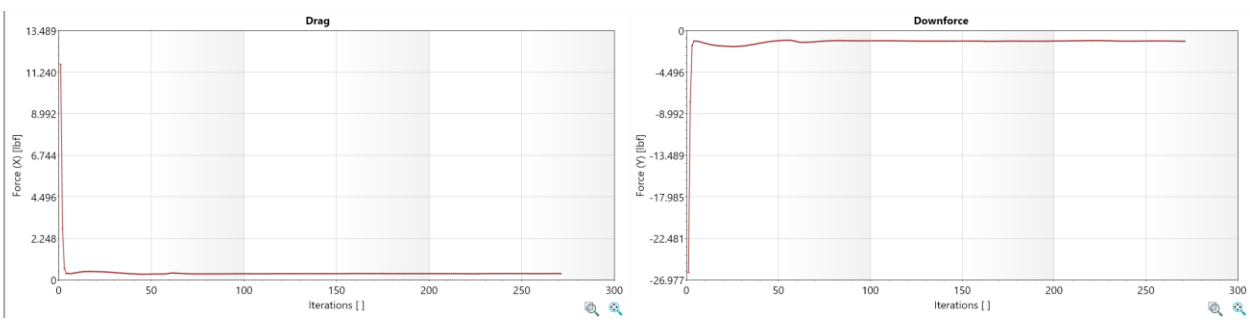

Figure 36. Cutouts, 10 Degrees Angle of Attack, Drag and Lift Forces vs. Iterations

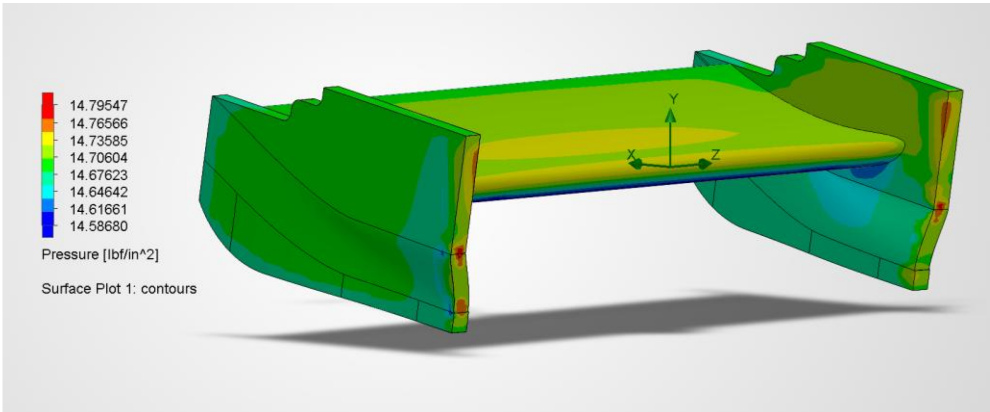

Figure 37. Cutouts, 10 Degrees Angle of Attack, Pressure Surface Plot

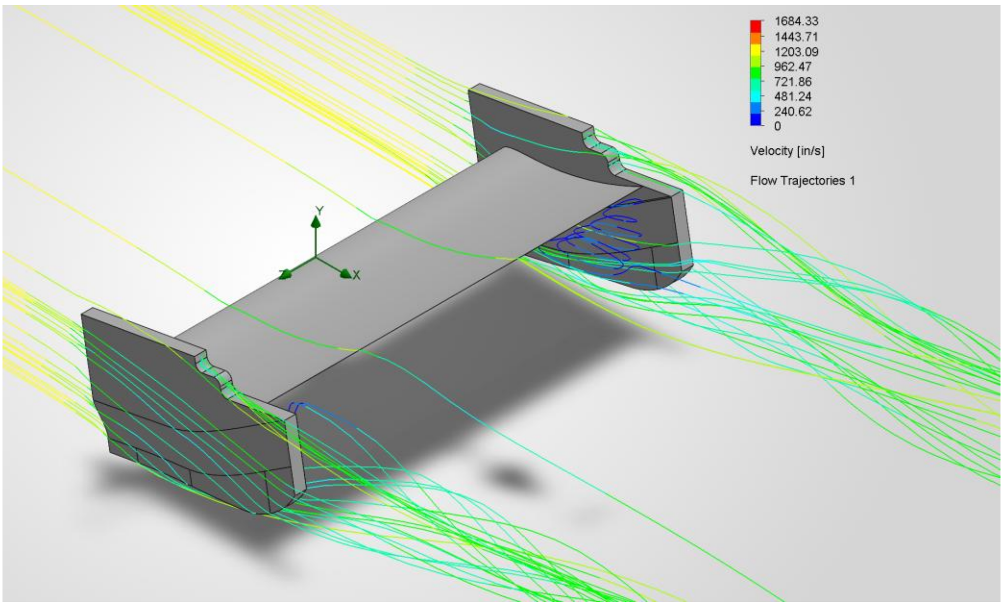

Figure 38. Cutouts, 10 Degrees Angle of Attack, Velocity Flow Trajectories

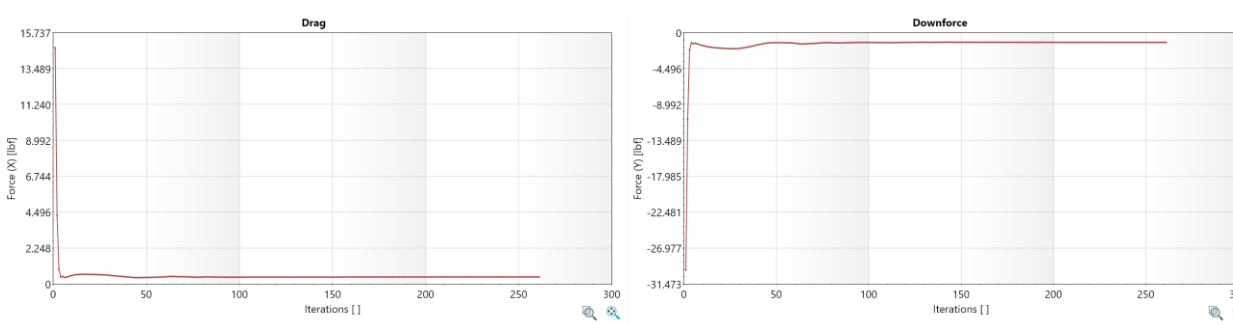

Figure 39. Cutouts, 15 Degrees Angle of Attack, Drag and Lift Forces vs. Iterations

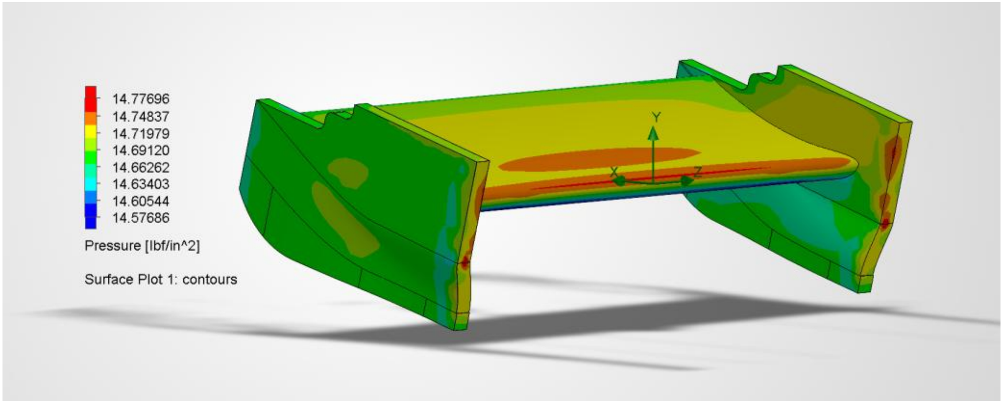

Figure 40. Cutouts, 15 Degrees Angle of Attack, Pressure Surface Plot

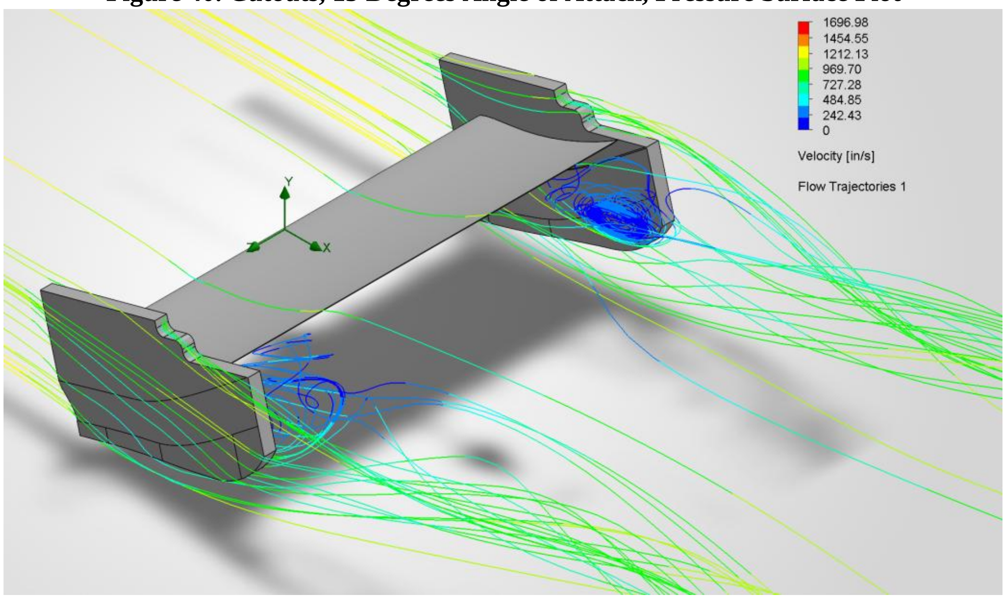

Figure 41. Cutouts, 15 Degrees Angle of Attack, Velocity Flow Trajectories

All four rear wing endplate geometries demonstrated convergence throughout the CFD simulations, with the goal plots for both lift and drag only experiencing minor fluctuations during the initial iterations. The convergence of each of these different types of endplates allows for a direct comparison of their aerodynamic performance. The reference, strakes, and cutout geometries all had a similar number of cells, while the slats required a larger amount to account for the airflow in through the specific geometry. As the angle of attack increased from 0° to 15°, each geometry experienced an increase in downforce as well as drag due to the larger pressure differences across the rear wing surfaces. The pressure contours remained relatively similar among the four endplate designs, the high pressure regions on the upper surface of the airfoil and the low pressure regions beneath became more pronounced as the angle of attack was increased contributing to the generation of more downforce. The velocity contours showed that each endplate modification altered the wake structure differently. The reference geometry produced progressively stronger wingtip vortices as the angle of attack increased, the strake geometry generated stronger streamwise vortices the remained attached further downstream. The slat geometry redirected airflow through the openings produce to a more organized wake, and the cutout geometry created secondary vortices near the trailing edge. These differences in flow behavior are reflected in the aerodynamic forces, with all configurations producing greater lift and drag at higher angles of attack. The strake geometry generated the greatest downforce at 15°, reaching -1.27 lbf with a corresponding drag of 0.46 lbf, while the slat geometry achieved the highest aerodynamic efficiency with a lift-to-drag ratio of approximately 4.10 at 10°. These results demonstrate that relatively small modifications to the rear wing endplate significantly influence pressure distribution, vortex formation, and wake development, allowing each geometry to provide a different balance between maximum downforce and aerodynamic efficiency depending on the desired performance objective.

IV. Prototype Building and Testing

The manufacturing process began by creating the baseline airfoil and a reference endplate in SolidWorks. Using the reference endplate, each group member designed a different endplate geometry to investigate how changes in endplate shape affect lift, drag, and downforce. Once all the designs were finalized, the parts were 3D printed with black PLA filament (fig. 42-45). Since the prototypes were going to be tested in the wind tunnel, the additive manufacturing lab technician was notified so that the print settings, including the infill density, could be adjusted for the application. While the printed parts closely matched the CAD models, the surface finish was rougher than expected due to the printing process. Unfortunately, the endplates with louvers did not print correctly, and due to time constraints, they were mocked up as a reference design for the endplates. Electrical tape was used to cover up the holes, and these manufacturing imperfections were considered as possible sources of uncertainty during the wind tunnel testing and when comparing the experimental results to the CFD analysis.

Figure 42. Printed Reference Endplate

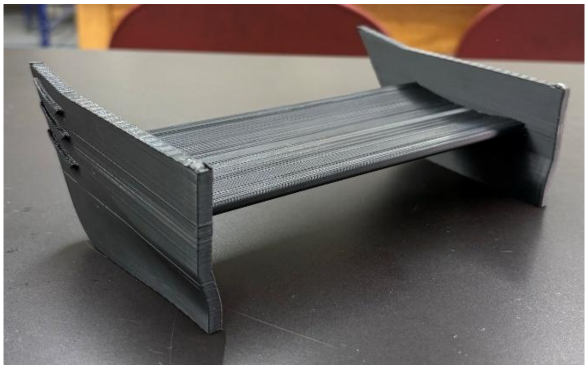

Figure 43. Printed Strakes Endplate

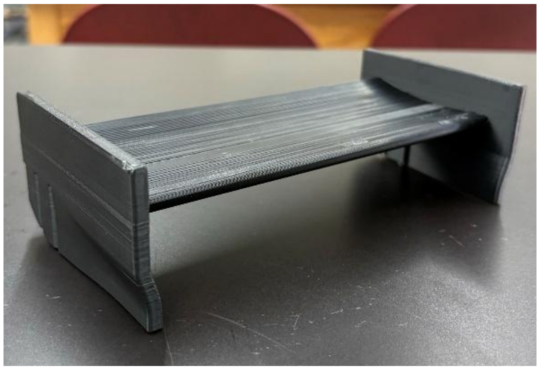

Figure 44. Printed Slats Endplate

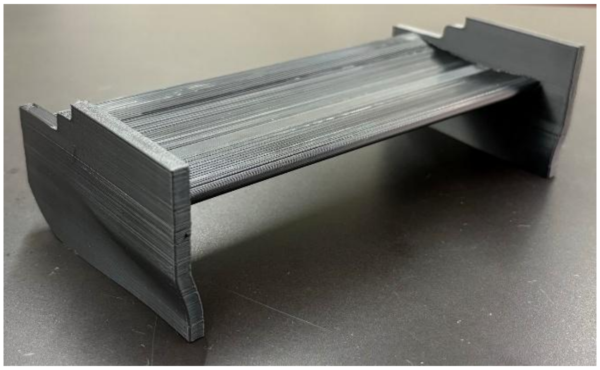

Figure 45. Printed Cutouts Endplate

The wind tunnel testing was carried out via three tests for each of the four wings, with each test having different angles of attack. The wings were secured in the wind tunnel by heat-pressing a metal thread into the bottom of each wing so that a metal rod could be screwed to the bottoms of the wings. This metal rod was then secured in the bottom of the wind tunnel, and the added effect of the support rod on drag was considered. The support rod allowed for the setting of different angles of attack, being 0°, 10°, and 15°. Data was recorded at 100% velocity capable by the wind tunnel, with 100% velocity being approximately 1181.1in/s. Due to limited time available in the wind tunnel lab, only one run per angle of attack of each wing was allowed. To further improve accuracy, multiple runs for each angle of attack would be beneficial to get an average that reinforces the difference in drag and lift as result of changing angles of attack. An attempt to create a smoke-slipstream was made, but the results were not worth documenting and a dedicated wind tunnel with built in smoke streams is needed. Of the data recorded (Fig. 46-47), lift and drag are the primary focus of the testing, with the percent aerodynamic efficiency calculated as lift over drag. Charts were made to make a visual comparison of the recorded data (Fig. 48-49), with the graph of aerodynamic efficiency (Fig. 50) being the most descriptive of how angle of attack combined with different geometry effects the lift and drag of rear wings.

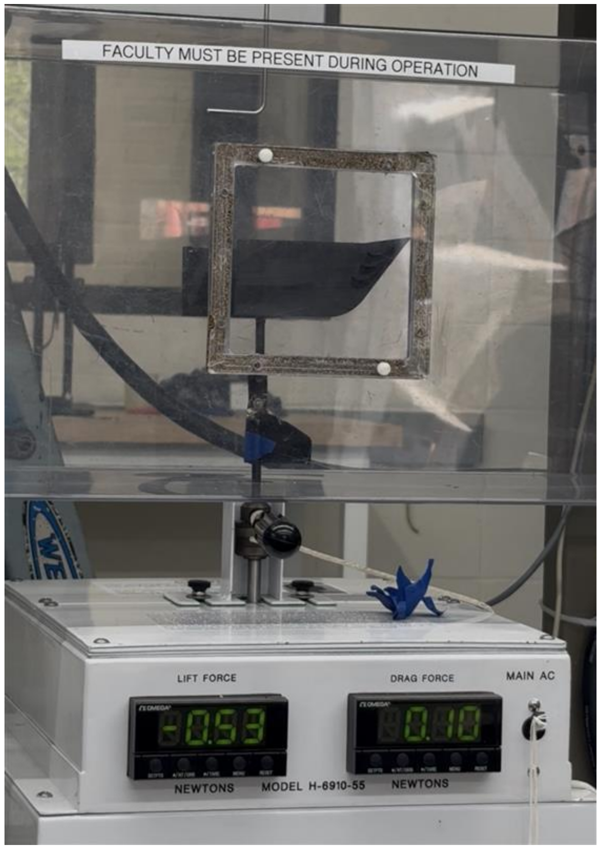

Figure 46. Strakes Model in Wind Tunnel at 0 Degrees Angle of Attack

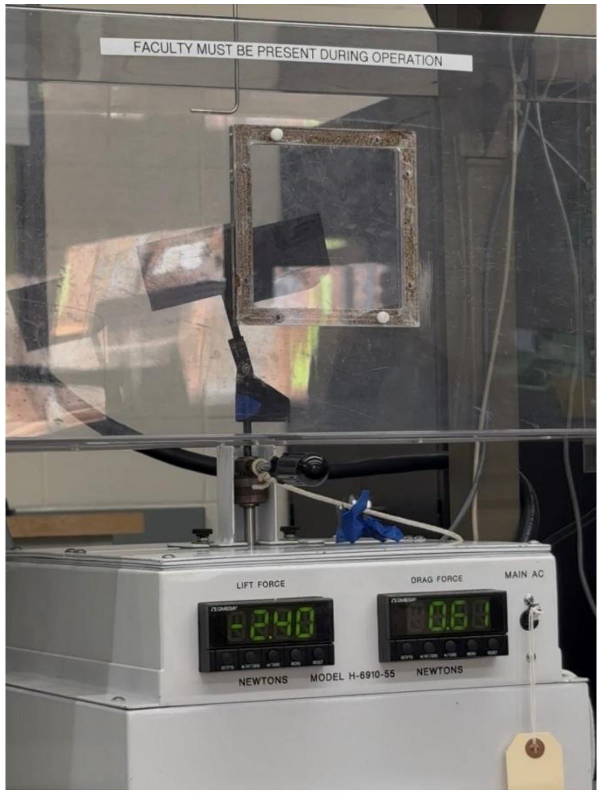

Figure 47. Reference Model in Wind Tunnel at 10 Degrees Angle of Attack

100% Velocity			
Reference			
Angle of Attack (deg)	Lift (N)	Drag (N)	% Aerodynamic Efficiency (Lift/Drag)
0	-2.15	0.54	3.98
10	-3.20	0.88	3.64
15	-3.91	1.17	3.34
Slats			
Angle of Attack (deg)	Lift (N)	Drag (N)	% Aerodynamic Efficiency (Lift/Drag)
0	-1.69	0.50	3.38
10	-3.32	0.81	4.10
15	-4.14	1.08	3.83
Cutouts			
Angle of Attack (deg)	Lift (N)	Drag (N)	% Aerodynamic Efficiency (Lift/Drag)
0	-1.90	0.65	2.92
10	-3.69	0.95	3.88
15	-4.20	1.19	3.53
Strakes			
Angle of Attack (deg)	Lift (N)	Drag (N)	% Aerodynamic Efficiency (Lift/Drag)
0	-2.23	0.54	4.13
10	-2.78	0.84	3.31
15	-4.47	1.27	3.52

Table 4. Maximum Velocity Data

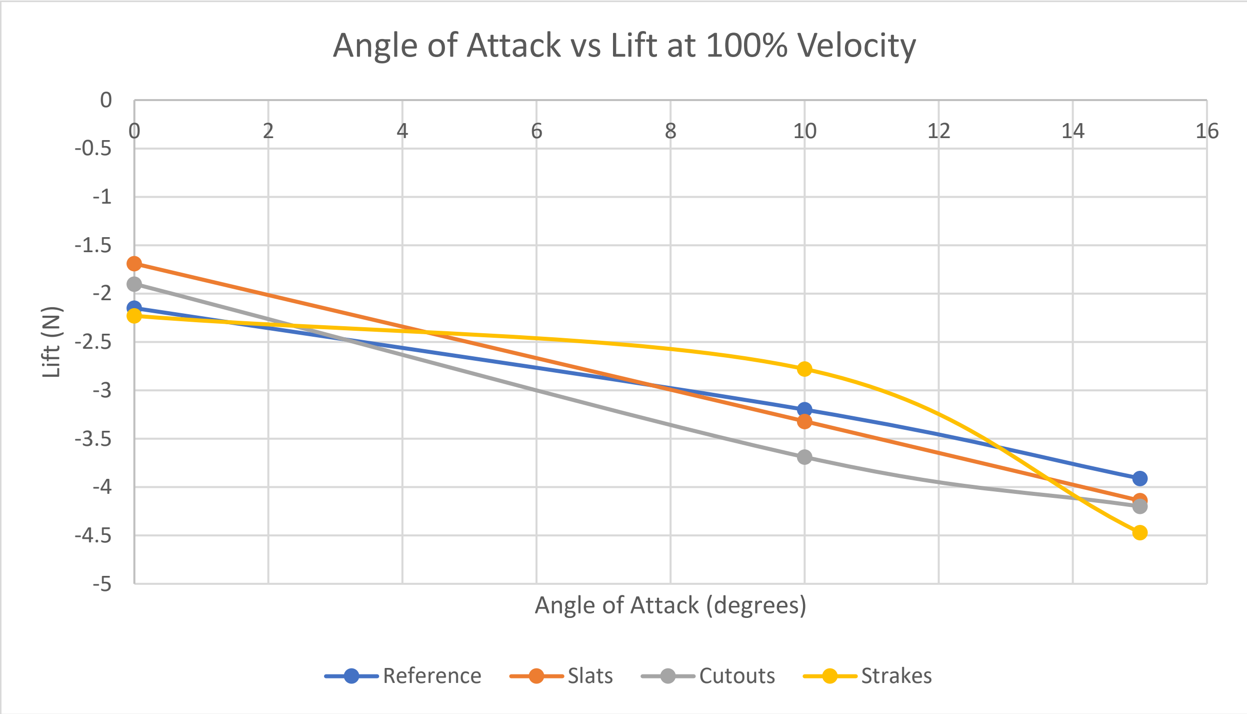

Figure 48. Angle of Attack vs. Lift at 100% Velocity

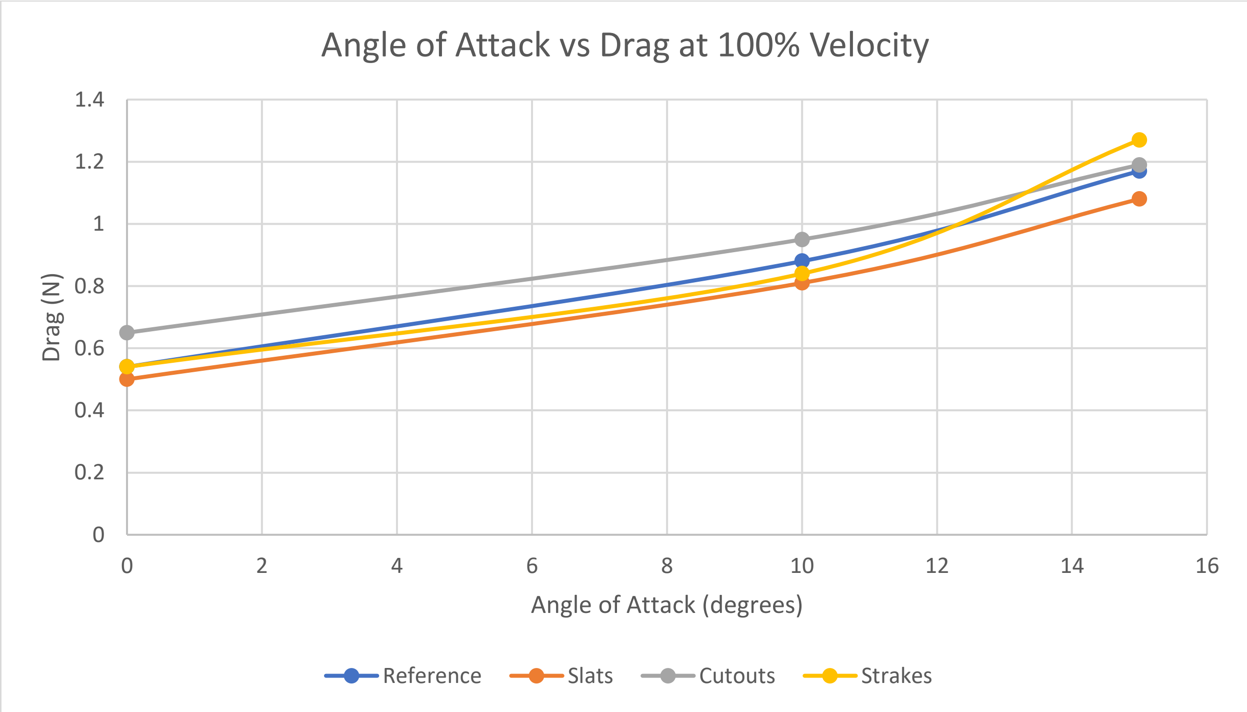

Figure 49. Angle of Attack vs. Drag at 100% Velocity

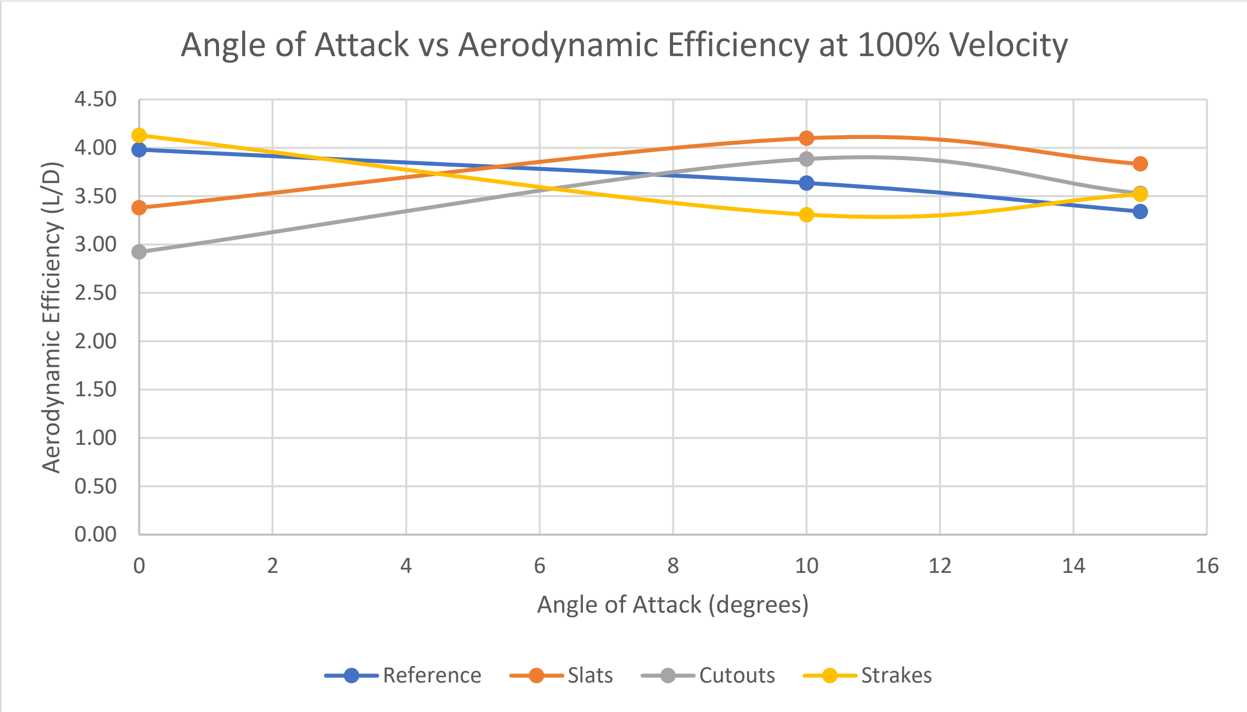

Figure 50. Angle of Attack vs. Aerodynamic Efficiency at 100% Velocity

V.Validation

To further validate the CFD values, the lift and drag coefficients were calculated for the CFD *and* experimental recorded data to express the measured forces in dimensionless form and facilitate comparison between the CFD and experimental results. This is because there may be a large discrepancy between the CFD forces and experimental forces, so a comparison of the forces alone would lead to much larger error percentages. The theory is that the lift and drag coefficients should be closer in value for CFD and the experiment because it brings context (i.e. area, and air velocity & density) to the values for comparison's sake. The air velocity is the same velocity used in the CFD simulations and experimental air tunnel. The projected area is the area from the top view of each model, and each model's projected area stays constant regardless of angle of attack. The frontal area is the area that is perpendicular to the velocity, and each model's frontal area changes with the angle of attack. Finally, the air density is taken from the air temperature of the CFD simulation.

Lift Coefficient:

$$\boldsymbol{C_L = \frac{L}{\frac{1}{2}\rho V^2 A_P}}$$

Example: Reference Model at 0° Angle of Attack

$$\boldsymbol{C_L = \frac{-0.483\ lbf}{\left(\frac{1}{2}\right)\left(0.002445\ \frac{slug}{ft^3}\right)\left(98.4252\ \frac{ft}{s}\right)^2 (0.1231\ ft^2)}}$$

$$\boldsymbol{C_L = -0.3315}$$

Drag Coefficient:

$$C_D = \frac{D}{\frac{1}{2}\rho V^2 A_f}$$

Example: Reference Model at 0° Angle of Attack

$$C_D = \frac{0.1214\ lbf}{\left(\frac{1}{2}\right)\left(0.002445\frac{slug}{ft^3}\right)\left(98.4252\frac{ft}{s}\right)^2 (0.0330in^2)}$$

$$C_D = 0.3116$$

Tables 5 and 6 showcase the CFD and experimental values respectively. In both Table 5 and 6, each model generates roughly the same amount of lift and drag when tilted to the same angle of attack during their respective velocity runs. Table 5 shows lift coefficients values that range from roughly -0.5 to -0.9, with the values increasing as the angle of attack increased. However, Table 6 shows lift coefficients that were almost always 0.2 greater than the values from Table 5, ranging from values of -0.3 to -0.7. Though each experimental model's lift coefficient increased as the angle of attack increased, like the CFD simulations. Further, the CFD drag coefficients follow the same pattern as the CFD lift coefficients; the values are consistently increasing as the angle of attack increases for all model configurations, but at a lesser pace than the lift coefficients. Similarly, the experimental drag coefficients do increase with the angle of attack, but at a much smaller range than the lift coefficients. The CFD drag coefficients range from values of 0.5 to 0.66, and the experimental drag coefficients range from values of 0.3 to 0.4.

CFD				
0 Degrees				
Model	Lift (lbf)	Drag (lbf)	C_L	C_D
Reference	-0.72	0.20	-0.49074	0.5063
Slat	-0.69	0.20	-0.47112	0.52153
Strakes	-0.75	0.21	-0.52226	0.50496
Cutout	-0.73	0.19	-0.50406	0.49762
10 Degrees				
Model	Lift (lbf)	Drag (lbf)	C_L	C_D
Reference	-1.12	0.35	-0.76986	0.607
Slat	-1.14	0.35	-0.77979	0.61503
Strakes	-1.21	0.37	-0.84459	0.6102
Cutout	-1.14	0.34	-0.77882	0.60203
15 Degrees				
Model	Lift (lbf)	Drag (lbf)	C_L	C_D
Reference	-1.21	0.43	-0.82643	0.6545
Slat	-1.21	0.43	-0.8329	0.65463
Strakes	-1.27	0.46	-0.88633	0.65715
Cutout	-1.22	0.43	-0.83666	0.66134

Table 5. Values Obtained from CFD

Experimental				
0 Degrees				
Model	Lift (lbf)	Drag (lbf)	C_L	C_D
Reference	-0.48	0.12	-0.33149	0.31164
Slat	-0.38	0.11	-0.26058	0.28858
Strakes	-0.50	0.12	-0.35072	0.29521
Cutout	-0.43	0.15	-0.29292	0.37484
10 Degrees				
Model	Lift (lbf)	Drag (lbf)	C_L	C_D
Reference	-0.72	0.20	-0.49335	0.34755
Slat	-0.75	0.18	-0.51185	0.31991
Strakes	-0.62	0.19	-0.43722	0.31410
Cutout	-0.83	0.21	-0.56889	0.38020
15 Degrees				
Model	Lift (lbf)	Drag (lbf)	C_L	C_D
Reference	-0.88	0.26	-0.60281	0.39726
Slat	-0.93	0.24	-0.63827	0.36672
Strakes	-1.00	0.29	-0.70302	0.40841
Cutout	-0.94	0.27	-0.64752	0.41123

Table 6. Values Obtained from Wind Tunnel Testing

The differences between the CFD and experimental results, as seen in Table 7, can be attributed to several sources of uncertainty, and while limited access to the wind tunnel allowed only one functional test to be completed for each endplate geometry and angle of attack, this prevented multiple trials from being conducted and eliminated the opportunity to modify or optimize the designs based on the initial experimental results. Ongoing construction outside the fluids laboratory also significantly limited access to the wind tunnel, which had only been available for a few days of testing and had undergone minimal recent use, as a result, calibration uncertainty may have influenced the measured forces. The 3D-printed endplates likewise did not produce the smooth surface finish originally intended, and although surface roughness was incorporated into the CFD model in an effort to better represent the printed parts, it remains uncertain whether this fully captured their aerodynamic behavior. Additional discrepancies may have resulted from small mounting and alignment tolerances during testing, evident of the simplified assumptions inherent in the CFD simulations carrying their own share of the disagreement between the two methods.

Percent Error				
0 Degrees				
Model	Lift	Drag	C_L	C_D
Reference	32.45	38.45	32.45	38.45
Slat	44.69	44.66	44.69	44.66
Strakes	32.84	41.54	32.84	41.54
Cutout	41.89	24.67	41.89	24.67
10 Degrees				
Model	Lift	Drag	C_L	C_D
Reference	35.92	42.74	35.92	42.74
Slat	34.36	47.98	34.36	47.98
Strakes	48.23	48.52	48.23	48.52
Cutout	26.95	36.85	26.95	36.85
15 Degrees				
Model	Lift	Drag	C_L	C_D
Reference	27.06	39.30	27.06	39.30
Slat	23.37	43.98	23.37	43.98
Strakes	20.68	37.85	20.68	37.85
Cutout	22.61	37.82	22.61	37.82

Table 7. Error Between CFD and Experimental

The CFD and wind-tunnel results showed partial qualitative agreement, particularly in predicting increased aerodynamic loading with increasing angle of attack. However, differences in force magnitudes and configuration rankings prevented quantitative validation of the numerical model. These two sets of data showed percent errors ranging from 20.68% to 48.52%. This quantitative difference is primarily driven by the computer simulation assumed a perfectly clean airflow environment. It did not include the physical mounting fixture and metal support rod used to hold the models inside the actual wind tunnel. Additionally, only one run in the wind tunnel was made for each specimen, thus producing inaccurate values for lift and drag. There were also discrepancies in simulating the non-flat surface textures left behind during the 3D printing. While surface roughness was included to try and account for this, it is unlikely that the simulation values accurately portray this surface friction. The wind tunnel testing successfully confirms the qualitative accuracy of the CDF software. However, the simulation fails to provide true quantitative validation.

VI. Conclusion

This study investigated how rear-wing endplate geometry influences downforce, drag, wingtip-vortex behavior, and overall aerodynamic efficiency. Four configurations: a reference endplate, slats, cutouts, and strakes were analyzed using computational fluid dynamics and experimentally evaluated using 3D-printed wind-tunnel models at angles of attack of 0°, 10°, and 15°. The CFD simulations provided visual and numerical insight into pressure distributions, velocity fields, and wake structures, while the wind-tunnel testing allowed the relative performance of each design to be compared under controlled physical conditions.

The experimental results demonstrated that no single endplate geometry performed best at every operating condition. At 0°, the strake configuration produced the greatest downforce and the highest downforce-to-drag ratio.

At 10°, the cutout configuration generated the greatest absolute downforce, while the slatted configuration produced the highest aerodynamic efficiency. At 15°, the strakes generated the greatest downforce at 4.47 N, but also produced the greatest drag. The slatted endplates generated 4.14 N of downforce with only 1.08 N of drag, resulting in the highest downforce-to-drag ratio at that angle. Compared with the reference geometry, the slatted configuration generated greater downforce and lower drag at both 10° and 15°. These results suggest that the slats effectively modified the endplate wake and provided the most favorable balance between aerodynamic loading and drag at the higher angles of attack.

Overall, the results confirm that relatively small changes in endplate geometry can significantly affect rear-wing performance and that the effectiveness of each feature depends on the operating angle of attack. The slatted design was the strongest overall configuration for applications prioritizing aerodynamic efficiency, while the strakes may be preferable when maximum downforce is more important than minimizing drag. The study also demonstrates the value of combining CFD with experimental testing, since simulations provide detailed flow visualization while wind-tunnel measurements verify the resulting aerodynamic forces. Future work should include repeated wind-tunnel trials, additional angles of attack and airflow velocities, further mesh-independence analysis, and testing with the rear wing integrated with a complete vehicle and diffuser. These improvements would provide a more complete understanding of how the endplate geometries perform under realistic racing conditions.

Acknowledgments

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